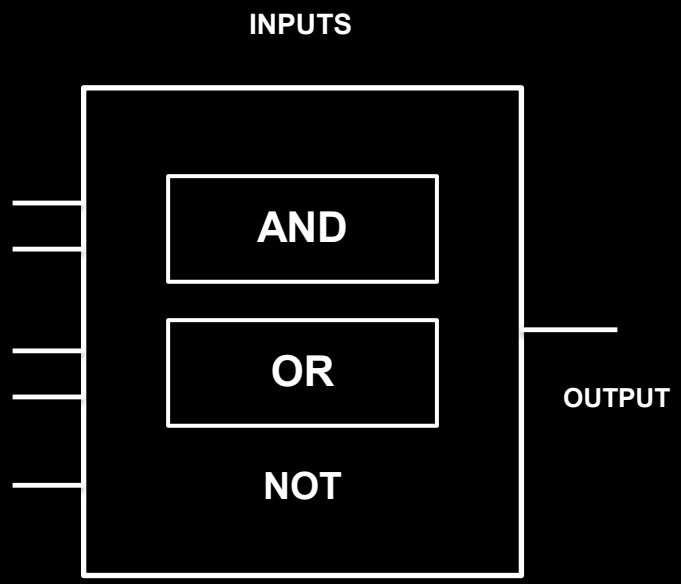
LOGIC

Truth tables • AND / OR / NOT • understand logic statements • operator precedence

Chapter 18 | Pearson Edexcel International GCSE Computer Science
Unit 4: Computers • textbook pages 181–186



Chapter 18: Logic

Two objectives guide the lesson.

OBJECTIVE

Use AND, OR and NOT gates; read and complete simple truth tables.

LESSON ROUTE

Conditions & Boolean values → AND / OR / NOT → truth tables → logic statements → precedence → checking logic

Chapter coverage follows the uploaded book pages 181–186, including worked examples, activities and checkpoint tasks.

Logic turns conditions into choices

A selection statement runs commands when its condition is TRUE.

THE IF PATTERN

```
IF CONDITION THEN
    COMMANDS
END IF
```

CONDITION

A test that must be satisfied before a command runs.

COMMAND

An instruction or order given to the computer.

BOOLEAN

A value that can only be TRUE or FALSE; often shown as 1 or 0.

Example: IF health <= 0 AND lives = 0 THEN game_over = true. The commands run only if the whole condition is TRUE.

Truth tables: test every possible case

A table records the inputs and the output for each combination.

WHAT A TRUTH TABLE SHOWS

- Every possible combination of inputs
- The output produced for each row
- Intermediate columns can show how a compound statement is worked out

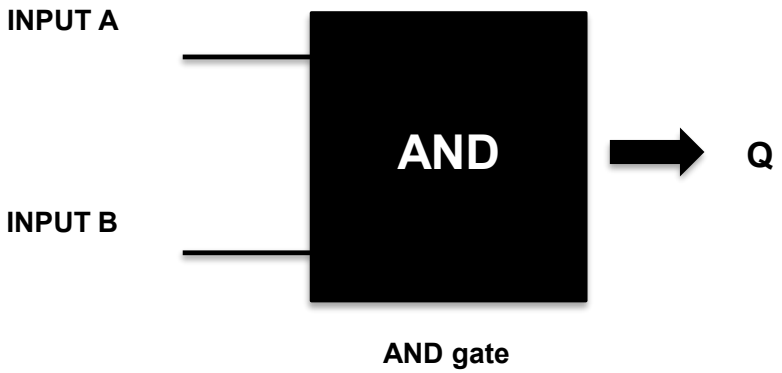
Two Boolean inputs create four combinations:

A	B	OUTPUT Q
0	0	?
0	1	?
1	0	?
1	1	?

Use 0 for FALSE / No and 1 for TRUE / Yes. Fill the output column only after deciding which operator is being tested.

AND: both conditions must be TRUE

Book example: the game ends when health is gone AND lives are gone.



AND gate

A = health <= 0 B = lives = 0

A	B	GAME OVER?
0	0	0
0	1	0
1	0	0
1	1	1

AND outputs 1 only on the row where both inputs are 1.

OR and NOT: two more operators

Use the wording of the rule to decide which operator is needed.

OR

Output is 1 if either input is 1 (or if both are 1).

Book example: health <= 0 OR score > 1,000,000 can set game_over to true.

A	B	A OR B
0	0	0
0	1	1
1	0	1
1	1	1

NOT

NOT reverses a Boolean value: 0 becomes 1 and 1 becomes 0. Unlike AND and OR, NOT has one input.

INPUT A	NOT A
0	1
1	0

AND, OR and NOT are logical / Boolean operators. Built electronically, they form logic circuits with inputs and outputs.

Build a compound truth table

Use intermediate columns so each operator is easy to check.

Q = (NOT G AND H) OR S

G = gold_mode H = health <= 0 S = score > 1,000,000

G	H	S	NOT G	NOT G AND H	GAME OVER Q
0	0	0	1	0	0
0	0	1	1	0	1
0	1	0	1	1	1
0	1	1	1	1	1
1	0	0	0	0	0
1	0	1	0	0	1
1	1	0	0	0	0
1	1	1	0	0	1

In words: game over if gold mode is off AND health is gone, OR if the score is over 1,000,000.